Part 3 – Structured Problem Decomposition

1. Initial Requirement Clarification

Before starting implementation, I would clarify a few important points to avoid rework later:

1. What format is the Excel file provided in (.xlsx or .csv)?
2. What exact fields are included in the file?
3. What is the new custom field (data type, required or optional, validation rules)?
4. How should duplicate customers be handled — merged, rejected, or updated?
5. In what format should the sales report be generated (PDF, Excel or should be shown in dashboard view)?
6. Should total sales include only confirmed orders or all orders?

2. Implementation Approach

A. Import 5,000 Customer Records

Step 1 –Analyze the Data

- Examine the Excel structure.
- Identify mandatory vs optional fields.
- Map columns to the existing database schema.

Step 2 – Data Validation

- Validate email format and required fields.
- Check for duplicates (by email or customer ID).
- Handle missing values.

Step 3 – Processing Strategy

Since 5,000 records is a moderate volume, I would:

- Process records in batches to avoid high memory usage.
- Use transactions to maintain consistency.
- Log failed records separately for review.

Step 4 – Import Summary

After processing:

- Display total records processed.
- Number successfully imported.
- Number failed with reasons.

B. Add Custom Field to Customer Profile

Step 1 – Database Change

- Add a new column with proper data type.
- Define whether it allows null values or requires a default.

Step 2 – Backend Update

- Update the customer model.
- Extend validation logic to include the new field.

Step 3 – API and UI Update

- Update create/update endpoints.
- Modify forms to include the new field.

Step 4 – Backward Compatibility

- Ensure existing customers are not affected negatively.
- Provide default values if required.

C. Generate Total Sales Report per Customer

Step 1 – Define Business Logic

- Confirm that only approved/confirmed orders are included.
- Group totals by customer.

Step 2 – Query Implementation

- Use database aggregation (SUM with GROUP BY).
- Add indexing if performance becomes an issue.

Step 3 – Output Format

- Provide downloadable Excel or PDF.
- Optionally allow dashboard visualization.

Step 4 – Performance optimization

- Use pagination for large datasets.
- caching if report is frequently requested.